Supplementary Material for InTraScale: Accelerating Video Generation via In-Trajectory Resolution Scaling

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Scope and Evidence Conventions

This supplement provides implementation details, protocol definitions, additional results, and paired statistical analyses for InTraScale. The main paper remains self-contained. All comparisons within a suite use matched text prompts, generation seeds, initial noise, scheduler, guidance, frame count, and output resolution.

Metric naming. We evaluate Subject Consistency, Background Consistency, Motion Smoothness, Aesthetic Quality, and Imaging Quality using the VBench custom-input protocol. We call their unweighted mean *VBench-5*. VBench-5 is a compact paper-specific summary and is not the official VBench Quality, Semantic, or Total score. Paired analyses use prompt–seed pairs as statistical units.

Method naming and provenance. The immutable experiment archives predate the final terminology. Internal case identifiers containing `talh`, `cll`, or `stage2` denote InTraScale, ITU-only, and ITU, respectively. The code-and-data package contains an explicit mapping and retains the internal identifiers so that archived results remain traceable.

Evidence coverage. The package includes validated Wan50 and Distill4 quality, timing, component, and paired-statistics exports. It also includes the checksum-verified 50-sample raw $368\times640 \rightarrow 720\times1248$ ITU operator evaluation and a deterministic aggregation script. The local archive does not contain the complete custom ITU/TTD checkpoints or exact hardware/software version metadata. These missing items are not reconstructed from filenames or earlier prose.

Implementation and Training Details

In-Trajectory Upsampler

ITU maps a 16-channel clean video latent to the target spatial grid without changing the temporal dimension. A 3D input projection and four spatiotemporal residual blocks precede the spatial lifting operator; four further residual blocks and a 3D output projection reconstruct the target latent. The hidden width is 256. For the $3/2$ grid ratio, a $3\times$ PixelShuffle is followed by anti-aliased stride-2 downsampling. For the approximately $2\times$ ratio, a $2\times$ PixelShuffle produces a 92×160 latent grid that is center-cropped to 90×156.

ITU minimizes

$$\mathcal{L}_{\mathrm{ITU}} = \mathcal{L}_{\mathrm{rec}} + 0.2\mathcal{L}_{\mathrm{lf}} + 0.1\mathcal{L}_{\mathrm{temp}},$$

where the reconstruction term is Charbonnier loss with $\epsilon = 10^{-3}$, the low-frequency term preserves downsampled content, and the temporal term matches first-order latent differences.

Setting	ITU	TTD
Optimizer	AdamW	AdamW
Maximum steps	50,000	10,000
Precision	bf16	bf16
Learning rate	1×10^{-4}	5×10^{-5}
Weight decay	0.01	0.01
Effective batch size	8	8
Gradient clipping	1.0	1.0
Training seed	1234	1234
EMA	0.9999	none
Validation fraction	0.05	0.02
Trainable parameters	ITU only	rank-32 LoRA only

Table 1: Final training settings encoded in the supplied configurations. ITU uses batch size 1 with distributed ranks/accumulation totaling an effective batch size of 8.

Trajectory-Tail Distillation

TTD is a transition-local LoRA update. It targets the attention q, k, v, o projections and the two feed-forward projections, uses rank and training scale $r = \alpha = 32$, and keeps the base denoiser frozen. Its loss is

$$\mathcal{L}_{\mathrm{TTD}} = \mathcal{L}_1 + 0.1\mathcal{L}_{\mathrm{MSE}} + \lambda_{\mathrm{tt}}\mathcal{L}_{\mathrm{temp}}.$$

We use $\lambda_{\mathrm{tt}} = 0.05$ for Wan50 step 40 and zero for Wan50 step 45 and Distill4 step 3. The inference strength is selected from $\{0.25, 0.5, 0.75, 1.0\}$ on held-out development prompts; 0.75 is then fixed for the test sets.

Complete final-parameter inventory. Table 1 gives the shared optimization settings. ITU has 16 input/output channels, hidden width 256, eight residual blocks, zero dropout, and no residual bypass. Its main 368×640 route uses the

$2\times$ PixelShuffle-and-crop operator; the 480×832 operator study uses the rational $3/2$ operator. The remaining ITU loss weights are $(\lambda_{\rm rec}, \lambda_{\rm lf}, \lambda_{\rm temp}, \lambda_{\rm res}) = (1, 0.2, 0.1, 0)$. TTD uses batch size one per rank, rank and α equal to 32, targets $q, k, v, o, \text{ffn.0}, \text{ffn.2}$, uses L_1 and MSE weights 1 and 0.1, and has zero velocity-loss weights. Checkpoints are written every 1,000 ITU steps and 500 TTD steps. The machine-readable inventory in the code-and-data package additionally lists every generation, scheduler, attention, guidance, resolution, seed, and timing parameter for each suite.

Development ranges and selection. The development variables, and no others, were swept for final protocol selection. Wan50 transition steps $\{30, 35, 40, 45, 50\}$ were inspected under matched prompt–seed generation; steps 40 and 45 are both reported to expose the quality–efficiency trade-off. Wan50 step-40 TTD strength was evaluated at $\{0.5, 0.75, 1.0\}$. Distill4 evaluated the Cartesian product of TTD strength $\{0.25, 0.5, 0.75, 1.0\}$ and reconstruction noise $\{\text{random}, \text{resized LR flow}\}$ on eight validation-only prompts. The rule was to maximize validation VBench-5; exact ties prefer lower strength and then lexicographic noise mode. This selected strength 0.75 with random target-grid noise. Endpoint baselines report, rather than select among, all refinement counts $\{0, 1, 2, 4\}$ and all three resizers {latent interpolation, ITU, RGB Real-ESRGAN}. Architecture, optimizer, loss, LoRA-rank, guidance, and scheduler settings were fixed before these recorded sweeps and were not chosen on the final test prompts.

Evaluation Protocol

Generation. Wan50 uses Wan2.1 T2V-1.3B with 50 denoising evaluations and transition steps 40 or 45 for InTraScale. Distill4 uses Wan2.1 T2V-14B StepDistill-CfgDistill with scheduler steps $[1000, 750, 500, 250]$; InTraScale-D4 performs three low-resolution evaluations and one target-resolution evaluation. Every output contains 81 frames at 720×1248.

Prompts and randomness. Quality uses ten matched prompt–seed pairs in each suite. Wan50 uses seeds 9700–9709 and Distill4 uses seeds 9800–9809. The state-reconstruction and strength selection study uses eight disjoint prompts with seed base 16000. Prompt lists are included in the code-and-data package.

Metrics and motivation. Let $x, \hat{x} \in [0,1]^N$ denote matched decoded RGB videos. We define $\text{MSE} = N^{-1}\|x - \hat{x}\|_2^2$ and $\text{PSNR} = 10\log_{10}(1/\text{MSE})$; PSNR measures absolute reconstruction fidelity. SSIM compares local luminance, contrast, and structure and therefore complements pixel error. LPIPS is the mean learned feature distance between corresponding decoded frames and measures perceptual rather than exact-pixel agreement. Latent L_1 is the elementwise mean absolute difference between predicted and target clean latents. Temporal L_1 is $N_\Delta^{-1}\|\Delta_t x - \Delta_t \hat{x}\|_1$, measuring whether first-order frame evolution is preserved. HF-energy error is the absolute difference between mean high-pass residual energies after removing the low-pass component, measuring detail-statistics mismatch.

For generated-video quality, VBench-5 is

$$\tfrac{1}{5}(S_{\rm subj} + S_{\rm back} + S_{\rm motion} + S_{\rm aesthetic} + S_{\rm imaging}).$$

The five terms jointly test subject persistence, background stability, smooth motion, perceptual appeal, and imaging/detail quality; their unweighted mean is a transparent paper-specific summary, not an official VBench aggregate. Temporal Consistency is also shown separately when diagnosing scheduler-state reconstruction. End-to-end latency is measured because the paper's primary claim is reducing generation cost while retaining quality; its precise inclusion boundary is given next.

Warm latency. The model and checkpoints remain resident in one process. We run one untimed warm-up and then five matched generations. CUDA synchronization surrounds each measurement. Pipeline latency covers denoising, TTD, ITU, scheduler state reconstruction, VAE decoding, and output writing; it excludes runner construction and checkpoint loading. Exact GPU, driver, CUDA, and framework versions are recorded by the recovered original-machine snapshot. It reports NVIDIA H100 80GB HBM3 GPUs (81,559 MiB each); the supplied training launchers use four ranks, while the later audit container exposed one GPU. The host had two Intel Xeon Platinum 8462Y+ sockets (32 physical cores per socket, 128 logical CPUs) and 2,159,612,928,000 bytes of RAM. It ran Ubuntu 22.04.5 LTS, kernel 5.14.0, NVIDIA driver 550.90.07, CUDA 12.8, cuDNN 9.10.2, Python 3.11.13, PyTorch 2.8.0+cu128, torchvision 0.23.0+cu128, Transformers 4.57.1, Diffusers 0.32.2, and Accelerate 1.11.0. The external framework revisions are LightX2V `c0fb09a827d8`, DiffSynth-Studio `6fbd3dd6526c`, and VBench `45e79ec14e69`; the code-and-data supplement records the full commits in a sanitized machine-readable summary with hashes of the underlying export records.

Statistics. For quality, we report prompt-paired means, nonparametric bootstrap 95% confidence intervals, win/loss counts, and two-sided sign tests. Timing variation is the sample standard deviation across the five warm repeats. These two forms of uncertainty are not interchanged.

Code, Data, and Media Inventory

The accompanying code-and-data ZIP contains the ITU model, TTD LoRA utilities, Wan50 and Distill4 bridges, data builders, training entry points, inference runners, VBench aggregation, paired-statistics scripts, final configurations, prompt lists, sanitized raw metric JSONs, and the compact CSV tables used here. The source snapshot includes all paper-specific preprocessing, construction, training, inference, and analysis modules together with a static internal-import closure check. A code-to-paper map and core-file comments point to the corresponding method subsections. The generated-data appendix records both construction targets and recovered realized manifests: Wan50 ITU and step-40/45 TTD each contain 1,000 records, and Distill4 ITU contains 5,000 records. It also records LMDB schemas, deterministic splits, prompt-hash counts, raw-video inventory sizes, and why ordinary public RGB video datasets cannot replace matched generator-internal states. The Distill4 step-3 TTD LMDB is explicitly marked for a corrected legacy-path follow-up rather than being inferred from its training output. The media ZIP contains two matched prompt–seed groups. Each group compares Trilinear, ITU-only, and InTraScale at 1248-by-720 resolution, 81 frames, and 16 frames per second. Legacy 640-by-368 files misleadingly named `Native-HR` are explicitly excluded.

Reproducibility boundary. Public base-model weights, large latent LMDBs, and generated-video test sets are not duplicated. More importantly, the current local archive does not contain the final custom ITU/TTD weights. The package therefore supports independent audit of implementation, protocols, raw metric exports, and statistical claims, but full regeneration requires adding the final custom checkpoints.

Additional Wan50 Results

ITU Cross-Resolution Reconstruction

TTD Endpoint Alignment

Factorial Separation of ITU and TTD

Additional Distill4 Results

Scheduler-State Reconstruction and Strength Selection

Method	LR evals	HR evals	VBench-5 ↑	Warm latency (s) ↓
Native-HR	0	50	0.82836	187.54 ± 0.26
Trilinear@40	40	10	0.77812	58.12 ± 0.20
InTraScale@40	40	10	0.80983	58.49 ± 0.16
Trilinear@45	45	5	0.76776	41.98 ± 0.19
RALU-style@45	45	5	0.80440	42.01 ± 0.19
InTraScale@45	45	5	0.80792	42.26 ± 0.20
Trilinear@48	48	2	0.76409	32.27 ± 0.18
ITU-0HR	50	0	0.79647	25.88 ± 0.11
ITU-1HR	50	1	0.80093	29.46 ± 0.22
ITU-2HR	50	2	0.80355	33.23 ± 0.10
ITU-5HR (LUVE-style)	50	5	0.80073	44.00 ± 0.27

Table 2: Wan50 warm-model operating points. Endpoint-first ITU baselines complete all 50 low-resolution evaluations before lifting and optional target-resolution refinement.

Input	Metric	Trilinear	ITU	Wins
$480p$	Latent L_1 ↓	0.19288	0.09913	49/50
$480p$	PSNR ↑	24.2927	29.9980	50/50
$480p$	SSIM ↑	0.71588	0.86961	49/50
$480p$	LPIPS ↓	0.23575	0.06294	50/50
$480p$	Temporal L_1 ↓	0.02121	0.01468	50/50
$480p$	HF-energy error ↓	0.00449	0.00161	48/50
$368p$	Latent L_1 ↓	0.24437	0.16284	50/50
$368p$	PSNR ↑	23.4238	24.1021	37/50
$368p$	SSIM ↑	0.67404	0.75920	48/50
$368p$	LPIPS ↓	0.33522	0.16403	50/50
$368p$	Temporal L_1 ↓	0.02388	0.02017	49/50
$368p$	HF-energy error ↓	0.00792	0.00233	50/50

Table 3: ITU operator evaluation on 50 clean-latent pairs per setting. “Wins” counts samples on which ITU is better. The raw 368p records and deterministic aggregation script accompany the submission.

Transition	Base L_1	TTD L_1	Reduction	95% CI of improvement	Wins
Wan50 $s = 40$	0.03215	0.02385	25.82%	[0.00479, 0.01302]	10/10
Wan50 $s = 45$	0.02363	0.01866	21.03%	[0.00260, 0.00840]	10/10
Distill4 $s = 3$	0.04286	0.04070	5.03%	[0.00148, 0.00299]	10/10

Table 4: Prompt-paired endpoint alignment at the selected TTD strength $\lambda = 0.75$. Positive confidence intervals indicate lower endpoint L_1 after TTD.

Suite	Effect	Before	After	Δ VBench-5
Wan50@40	ITU within matched transition	0.77812	0.80896	+0.03085
Wan50@40	TTD within ITU	0.80896	0.80983	+0.00087
Wan50@45	ITU within matched transition	0.76776	0.80842	+0.04066
Wan50@45	TTD within ITU	0.80842	0.80792	-0.00050
Distill4@3	ITU within matched transition	0.81796	0.85670	+0.03874
Distill4@3	TTD within ITU	0.85670	0.85680	+0.00010

Table 5: Factorial quality effects. ITU supplies the dominant aggregate perceptual gain; TTD targets transition endpoint mismatch rather than acting as a general perceptual enhancer.

Method	LR/HR	Subject	Background	Motion	Aesthetic	Imaging	VBench-5	Latency (s)
Native-HR4	0/4	0.97186	0.97557	0.99123	0.65203	0.71135	0.86041	42.49
Trilinear@3	3/1	0.95932	0.96937	0.98376	0.58954	0.58783	0.81796	18.33
InTraScale-D4@3	3/1	0.96226	0.97026	0.98815	0.64914	0.71418	0.85680	19.92
ITU-2HR (LUVE-style)	4/2	0.96238	0.97341	0.98683	0.65191	0.72450	0.85981	28.76
MrFlow-style	4/1	0.96379	0.97259	0.98796	0.65491	0.71591	0.85903	27.12

Table 6: Distill4 warm-model quality–efficiency. The last two methods are endpoint-first baselines and perform additional processing after completing the four-step low-resolution trajectory.

Baseline	Metric	Baseline–InTraScale	95% CI	Wins/Losses	p
Trilinear@3	VBench-5	-0.03884	$[-0.04840, -0.02921]$	0/10	0.00195
Trilinear@3	Temporal	-0.00264	$[-0.00434, -0.00130]$	1/9	0.02148
ITU-2HR	VBench-5	$+0.00301$	[0.00077, 0.00564]	7/3	0.34375
ITU-2HR	Temporal	-0.00333	$[-0.00547, -0.00152]$	1/9	0.02148
MrFlow-style	VBench-5	$+0.00223$	$[-0.00277, 0.00752]$	6/4	0.75391
MrFlow-style	Temporal	-0.00180	$[-0.00289, -0.00075]$	2/8	0.10938

Table 7: Distill4 prompt-paired statistics. Differences are baseline minus InTraScale; "Temporal" is temporal consistency, so higher is better. p is the two-sided sign-test value.

Target-grid noise	TTD strength	VBench-5	Temporal consistency	Selected
Seed-derived Gaussian	0.25	0.83816	0.97476	no
Seed-derived Gaussian	0.50	0.83893	0.97468	no
Seed-derived Gaussian	0.75	0.84009	0.97465	yes
Seed-derived Gaussian	1.00	0.83933	0.97464	no
Resized LR flow	0.25	0.81467	0.95862	no
Resized LR flow	0.50	0.81488	0.95853	no
Resized LR flow	0.75	0.81526	0.95860	no
Resized LR flow	1.00	0.81486	0.95859	no

Table 8: Eight-prompt development study. Scheduler-consistent target-resolution noise improves both summary quality and temporal consistency across all tested TTD strengths.

Reproducibility Checklist Notes

The standalone checklist uses conservative answers. This section records the evidence boundary behind every response that is not *yes*.

Theoretical contributions (2.1–2.8). The paper makes empirical and methodological contributions, not theorem-level theoretical contributions. The conditional items 2.2–2.8 are therefore left blank in the standalone checklist, as directed by the "If yes" condition.

Dataset usage (3.2–3.7). All training and evaluation samples are generated or constructed within the experiments using the frozen base generator and author-defined prompt–seed protocols; the paper does not draw a dataset from the existing literature. Model-endogenous supervision is selected because ITU must learn the VAE-specific cross-resolution correspondence and TTD requires matched states from the same generated trajectory, supporting *yes* for item 3.2. These intermediate training and evaluation collections are not introduced as a new dataset contribution, so items 3.3–3.6 are *NA*. Their construction is described and the code-and-data supplement includes the exact test and development prompt lists, dataset builders, sanitized metric exports, and 50-sample $368p$ operator records. The generated-data appendix now gives the construction targets, schemas, deterministic split algorithm, and the scientific reason conventional public video collections are not substitutes; item 3.7 is therefore *yes*. Large generated videos and latent LMDB tensors remain excluded because detailed description, rather than inclusion, is what item 3.7 requests.

Computational experiments (4.2–4.13). The supplement and machine-readable development record enumerate every parameter varied for final selection, the number and values tried, the selection rule, and which settings were fixed without a test-set search (4.2: *yes*). All paper-specific preprocessing, training, inference, evaluation, and analysis sources are supplied and pass the included internal-import closure check (4.3–4.4: *yes*). A public-release license has not been selected (4.5: *no*). Core implementation comments and the code-to-paper map cover each introduced method operation (4.6: *yes*). Seeds and sample counts are explicit; paired bootstrap confidence intervals and two-sided sign tests use prompt–seed pairs as units (4.7 and 4.10–4.12: *yes*). The recovered original-machine snapshot records the exact H100 SKU and memory, CPU and host memory, operating system, driver, CUDA/cuDNN, Python/PyTorch packages, and LightX2V, DiffSynth-Studio, and VBench revisions (4.8: *yes*). Formal definitions and motivations cover every reported metric, and the supplied final-parameter inventory covers architecture, loss, training, generation, scheduler, attention, randomness, resolution, and timing settings for both suites (4.9 and 4.13: *yes*).